

Linear Stability Analysis of Two Velocity Profiles using the Orr–Sommerfeld Equation

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1 Objective

The objectives are:

- Compute the most unstable eigenvalues for $\alpha = 0.20$ and $Re = 650$
- Compare wall-normal eigenfunctions $|\phi(y)|$
- Compare streamwise disturbance eigenfunctions
- Plot neutral stability curves in the (α, Re) plane

2 Introduction

Parallel shear flows may become unstable as the Reynolds number increases. Linear stability theory predicts a neutral boundary that separates stable and unstable regions. The Orr–Sommerfeld equation is used to determine whether disturbances grow or decay.

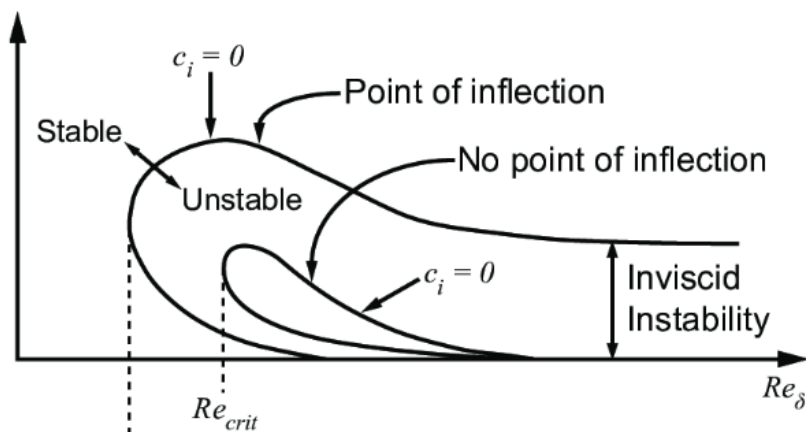


Figure 1: Typical neutral stability diagram separating stable and unstable regions.

3 Governing Equations

Small disturbances are assumed in the normal mode form

$$\psi(x, y, t) = \phi(y)e^{i(\alpha x - \omega t)}. \quad (1)$$

Substitution into the linearized Navier–Stokes equations gives the Orr–Sommerfeld equation:

$$(D^2 - \alpha^2)^2 \phi = i\alpha Re [(U - c)(D^2 - \alpha^2)\phi - U''\phi], \quad (2)$$

where $D = d/dy$, α is the wavenumber, Re is the Reynolds number, and $c = \omega/\alpha$ is the phase speed.

This forms the eigenvalue problem

$$A\phi = cB\phi. \quad (3)$$

$$\text{Im}(c) > 0 \Rightarrow \text{unstable}, \quad \text{Im}(c) < 0 \Rightarrow \text{stable}.$$

4 Numerical Method

Chebyshev spectral collocation was used to discretize the equation. Differentiation matrices were constructed, the generalized eigenvalue problem was solved, and the most unstable modes were extracted. Sweeping α and Re produced neutral stability curves.

5 Results

5.1 Most Unstable Eigenvalues ($\alpha = 0.20$, $Re = 650$)

Profile	Eigenvalue c	$\text{Im}(c)$	Stability
velocity1.dat	$0.999962 - 0.000998i$	< 0	Stable
velocity2.dat	$0.227375 + 0.103622i$	> 0	Unstable

Table 1: Comparison of most unstable eigenvalues.

5.2 Neutral Stability Curves

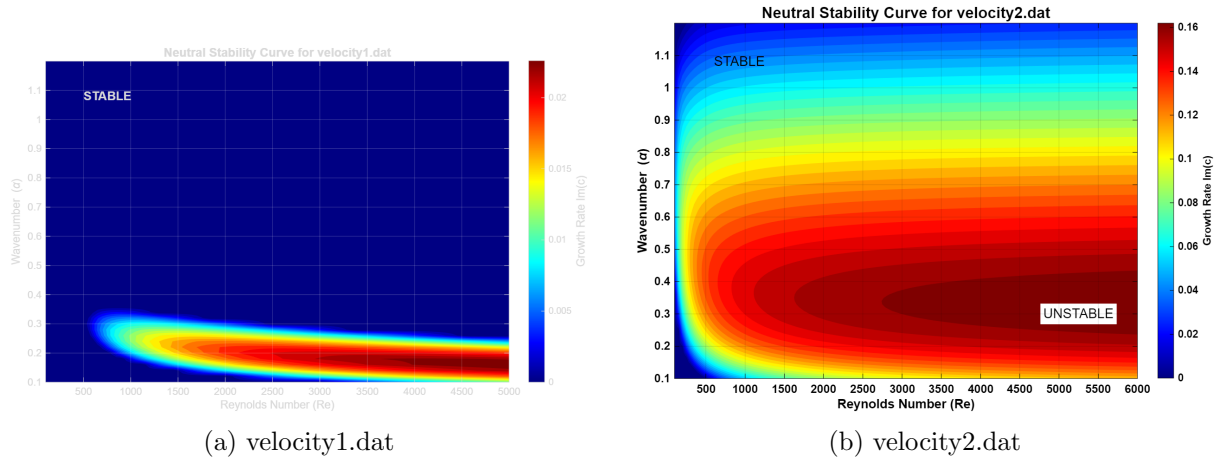


Figure 2: Side-by-side comparison of neutral stability contours.

Inference: Velocity2 shows a much larger unstable region and becomes unstable at lower Reynolds numbers, whereas velocity1 remains mostly stable. Hence, velocity2 is more susceptible to transition.

5.3 Wall-normal Eigenfunctions $|\phi(y)|$

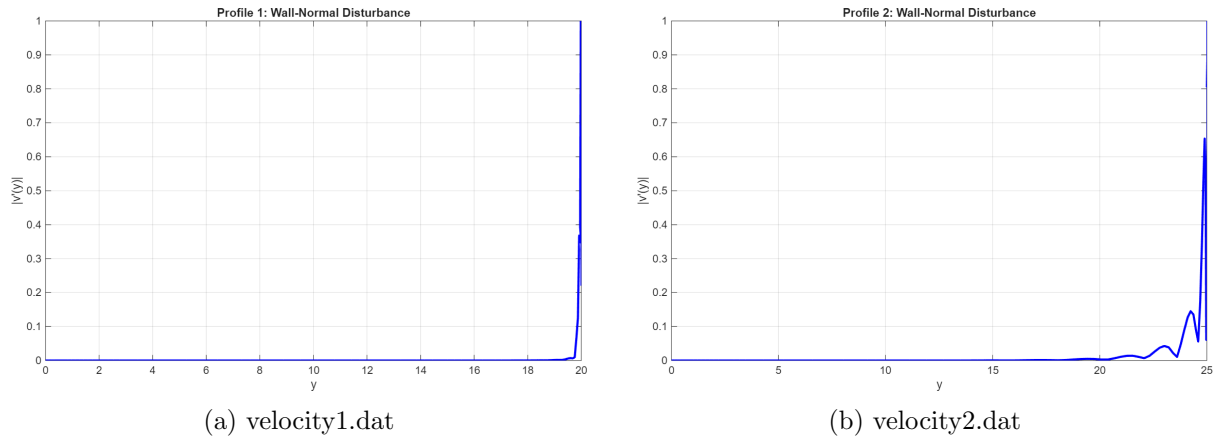


Figure 3: Wall-normal disturbance profiles.

Inference: Velocity2 exhibits stronger disturbance amplitudes near the shear region, indicating greater energy transfer from the base flow and enhanced instability.

5.4 Streamwise Eigenfunctions

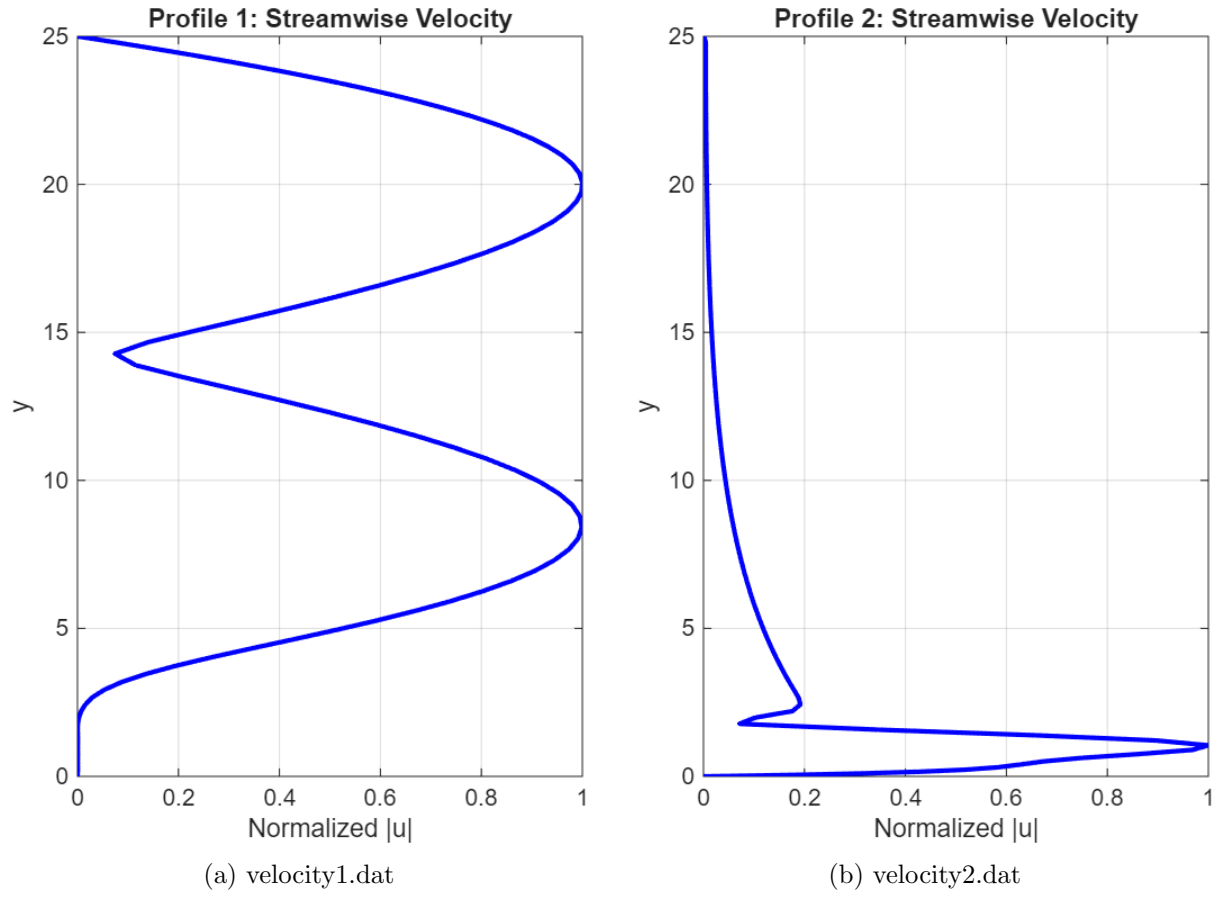


Figure 4: Streamwise disturbance profiles.

Inference: Larger variations in the streamwise disturbance for velocity2 confirm stronger amplification compared to velocity1.

6 Discussion

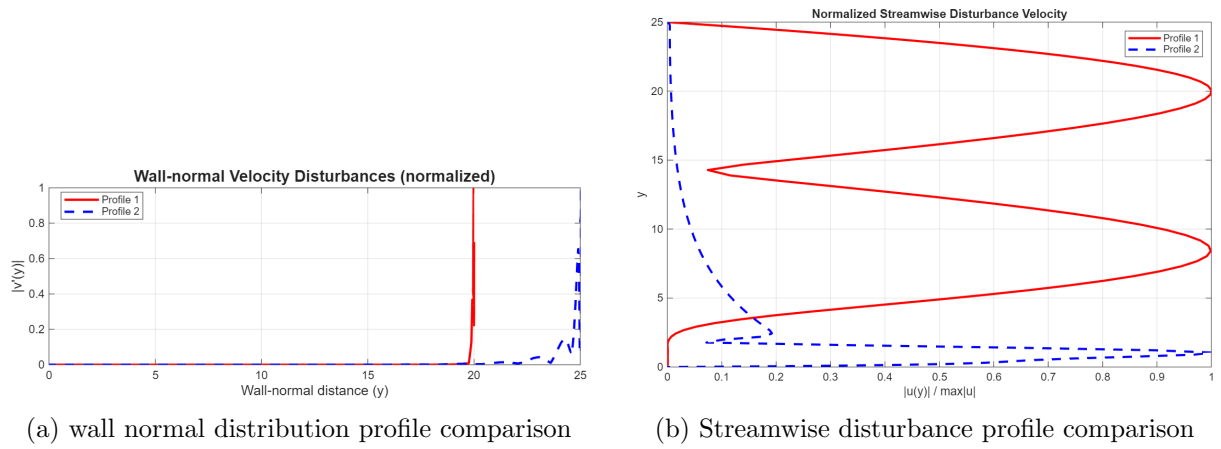
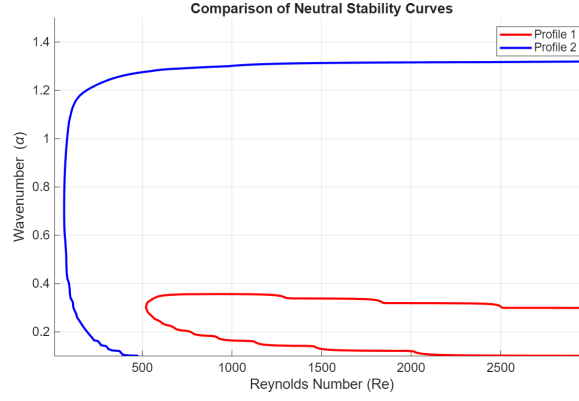


Figure 5: Comparison plots between Velocity1 and Velocity2.



(a) Comparison between neutral stability curves

The comparison clearly shows that velocity1 is stable while velocity2 is unstable for the same parameters. The difference in base flow shape strongly influences disturbance growth and transition behavior.

7 Conclusion

Orr–Sommerfeld stability analysis was performed for two velocity profiles. Velocity1 is stable at $\alpha = 0.20$ and $Re = 650$, while velocity2 is unstable.

8 Appendix: MATLAB Code

```
1 clear; clc; close all;
2 files = {'velocity1.dat', 'velocity2.dat'};
3 profile_name = {'Profile 1', 'Profile 2'};
4 alpha0 = 0.20;
5 Re0 = 650;
6 alpha_range = 0.1:0.05:1.5;
7 Re_range = 100:100:2000;
8 N = 99;
9 N1 = N+1;
10 j = (0:N)';
11 X = cos(pi*j/N); % Chebyshev nodes
12 phi_store = cell(2,1);
13 Y_store = cell(2,1);
14 u_store = cell(2,1);
15 v_store = cell(2,1);
16 for p = 1:2
17     % Load and map geometry
18     data = load(files{p});
19     ydat = data(:,1); Udat = data(:,2); U2dat = data(:,3);
20     L = ydat(end); scal = L/2; Y = X*scal + scal; fac = 1/scal;
21     Y_store{p} = Y;
22
23     % Interpolate base flow to Chebyshev grid
24     U = interp1(ydat, Udat, Y, 'pchip');
25     U2 = interp1(ydat, U2dat, Y, 'pchip');
26
27     % Physical differentiation matrices
28     [D0,D1,D2,~,D4] = Dmat_corrected(N);
29     D1_phys = fac*D1; D2_phys = fac^2*D2; D4_phys = fac^4*D4;
30
31     a1 = alpha0; Re = Re0; zi = 1i; a2 = a1^2; a4 = a2^2;
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32 B = (D2_phys - a2*D0);
33 A = (U*ones(1,N1)).*B - (U2*ones(1,N1)).*D0 - (D4_phys - 2*a2*D2_phys + a4*D0)/(zi*
    al*Re);
34
35 % Boundary conditions (v = 0, dv/dy = 0 at walls)
36 er = -200*zi;
37 A_bc = [er*D0(1,:); er*D1_phys(1,:); A(3:N-1,:); er*D1_phys(N1,:); er*D0(N1,:)];
38 B_bc = [D0(1,:); D1_phys(1,:); B(3:N-1,:); D1_phys(N1,:); D0(N1,:)];
39
40 % Solve and identify most unstable mode
41 [V,C] = eig(A_bc, B_bc);
42 cvals = diag(C);
43 [~, idx] = sort(imag(cvals), 'descend');
44
45 c_most = cvals(idx(1));
46 phi = V(:, idx(1));
47 u_dist = abs(D1_phys * phi);
48 v_dist = abs(-1i * alpha0 * phi);
49
50 % Normalize magnitudes
51 u_store{p} = u_dist / max(u_dist);
52 v_store{p} = v_dist / max(v_dist);
53
54 % Print complex eigenvalue
55 fprintf('%s (al=%.2f, Re=%d):\n', profile_name{p}, alpha0, Re0);
56 fprintf(' Full c = %.6f + %.6fi\n\n', real(c_most), imag(c_most));
57 % Individual disturbance plots
58 figure('Name', [profile_name{p} ' - u']);
59 plot(Y, u_store{p}, 'r-', 'LineWidth', 2);
60 xlabel('y'); ylabel('|u''(y)|'); title([profile_name{p} ': Streamwise']); grid on;
61 figure('Name', [profile_name{p} ' - v']);
62 plot(Y, v_store{p}, 'b-', 'LineWidth', 2);
63 xlabel('y'); ylabel('|v''(y)|'); title([profile_name{p} ': Wall-Normal']); grid on;
64 end
65 % Comparison plots
66 figure('Name', 'Comp - u');
67 plot(Y_store{1}, u_store{1}, 'r-', 'LineWidth', 2); hold on;
68 plot(Y_store{2}, u_store{2}, 'b--', 'LineWidth', 2);
69 legend(profile_name); xlabel('y'); ylabel('|u''(y)|'); title('Streamwise Comparison');
    grid on;
70 figure('Name', 'Comp - v');
71 plot(Y_store{1}, v_store{1}, 'r-', 'LineWidth', 2); hold on;
72 plot(Y_store{2}, v_store{2}, 'b--', 'LineWidth', 2);
73 legend(profile_name); xlabel('y'); ylabel('|v''(y)|'); title('Wall-Normal Comparison');
    grid on;
74 % Neutral stability maps
75 for p = 1:2
76     data = load(files{p});
77     ydat = data(:,1); Udat = data(:,2); U2dat = data(:,3);
78     L = ydat(end); scal = L/2; Y = X*scal + scal; fac = 1/scal;
79     U = interp1(ydat, Udat, Y, 'pchip'); U2 = interp1(ydat, U2dat, Y, 'pchip');
80     [D0,D1,D2,~,D4] = Dmat_corrected(N);
81     D1_phys = fac*D1; D2_phys = fac^2*D2; D4_phys = fac^4*D4;
82
83     ci_map = zeros(length(alpha_range), length(Re_range));
84     for ia = 1:length(alpha_range)
85         al = alpha_range(ia);
86         B = (D2_phys - al^2*D0);
87         for ir = 1:length(Re_range)
88             Re = Re_range(ir);
89             A = (U*ones(1,N1)).*B - (U2*ones(1,N1)).*D0 - (D4_phys - 2*al^2*D2_phys + al
                ^4*D0)/(1i*al*Re);
90             er = -200*1i;

```

```

91         A_bc = [er*D0(1,:); er*D1_phys(1,:); A(3:N-1,:); er*D1_phys(N1,:); er*D0(N1
, :)];
92         B_bc = [D0(1,:); D1_phys(1,:); B(3:N-1,:); D1_phys(N1,:); D0(N1,:)]];
93         eigvals = eig(A_bc, B_bc);
94         ci_map(ia,ir) = max(imag(eigvals));
95     end
96 end
97
98 figure('Name', ['Growth Map ' num2str(p)]);
99 [Re_grid, alpha_grid] = meshgrid(Re_range, alpha_range);
100 contourf(Re_grid, alpha_grid, ci_map, 20, 'LineStyle', 'none');
101 xlabel('Re'); ylabel('\alpha'); title(['Growth Rate Map: ', profile_name{p}]);
102 colorbar; grid on;
103 end
104 function [D0,D1,D2,D3,D4] = Dmat_corrected(N)
105     % Chebyshev differentiation using recurrence relations
106     N1 = N + 1; vec = (0:1:N)'; D0 = [];
107     for j = 0:1:N, D0 = [D0 cos(j*pi*vec/N)]; end
108     lv = length(vec);
109     D1 = [zeros(lv,1) D0(:,1) 4*D0(:,2)];
110     D2 = [zeros(lv,1) zeros(lv,1) 4*D0(:,1)];
111     D3 = [zeros(lv,1) zeros(lv,1) zeros(lv,1)];
112     D4 = [zeros(lv,1) zeros(lv,1) zeros(lv,1)];
113     for j = 3:N
114         D1 = [D1 2*j*D0(:,j) + j*D1(:,j-1)/(j-2)];
115         D2 = [D2 2*j*D1(:,j) + j*D2(:,j-1)/(j-2)];
116         D3 = [D3 2*j*D2(:,j) + j*D3(:,j-1)/(j-2)];
117         D4 = [D4 2*j*D3(:,j) + j*D4(:,j-1)/(j-2)];
118     end
119 end

```

9 References

Schmid, P. J., and Henningson, D. S., *Stability and Transition in Shear Flows*, Springer, 2001.